

Data engineer terms (youtube) (by ansh)

DATE:

1. Database:-

* It is structured data in the form of table (rows & columns)

* It follows ACID properties

Atomicity:-

* Transaction have to be completed entirely or completely roll back.

* Either happens or doesn't happen.

Consistency:-

* It have to follow constraints (rules)

ex: balance doesn't less than 0

Isolation:-

* Every person or client can able to work parallelly in a isolation

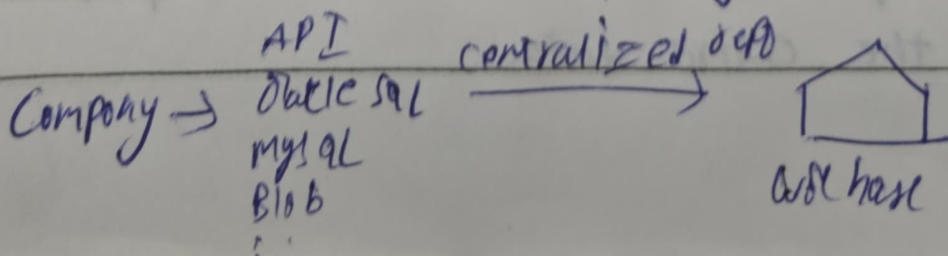
* Try with examples

Durability:-

* The result should be as expected.

2. Data warehouse:- * It store structured format.

* It is a centralized repository of a organization.



3. Data Lake :-

* It is used to store Structured, Semi Structured & Unstructured data.

4. Data Lakehouse :-

* It is going to add a layer to data lake to provide data warehouse capabilities.

5. Open table format :-

* It enables Data Lakehouse

* There are mainly 3 types of providers

1. Iceberg (2) Delta Lake

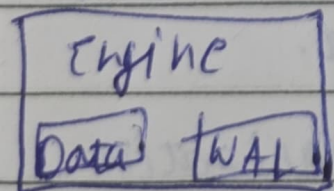
* Data Lake + Data Warehouse.

6. Data Lakebase :-

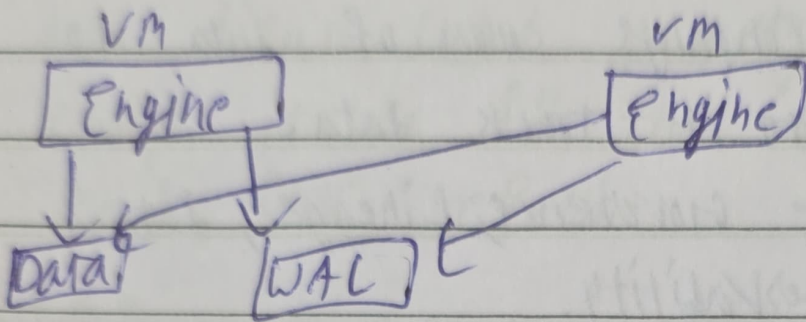
* It is built by databricks recently.

Previously

VM



* Data & WAL are dedicated to only one engine. If we want to use for another engine means we have to replicate the entire data.

Now with Lake base

- * Data & WAL are independent so no need to replicate the data.

7. Data mart:-

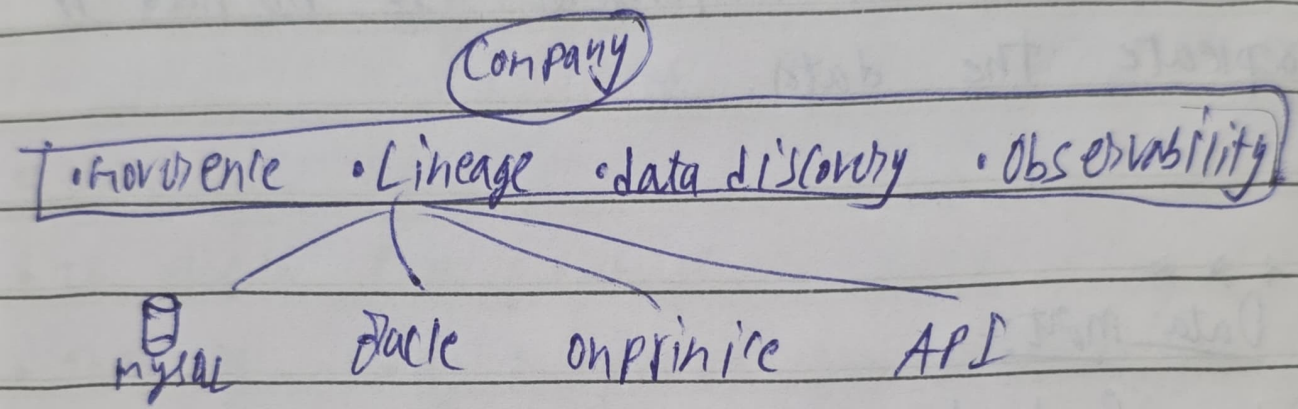
- * Subset of database or datawarehouse to dedicatedly particular use of one purpose.

8. Data mesh:-

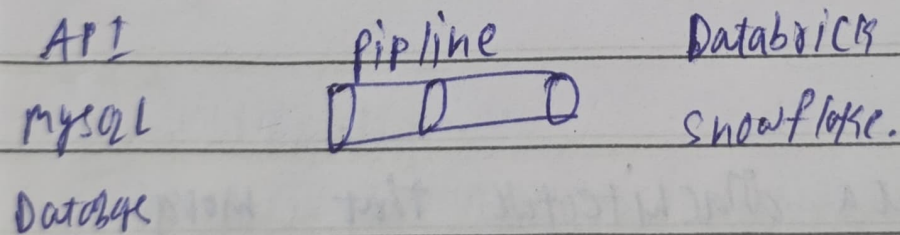
- * Decentralized data architecture that moves data ownership from a single central team to specific business domain teams.
- * Domain ownership → sales, hr, ... manages their own data.
- * Data as a product.

9. Data Fabric

- It is focus on company's source of data to provide a layer to it track data.
- That layer have Governance, Lineage, data discovery & observability.



10. ETL/ELT:-



11. Incremental data loading:-

- Data will be loaded incrementally.

12. Back Date refresh:-

- * After loading Incremental data, we find 2 days back data didn't loaded properly we have to change it so for that we have to enable Back Date refresh.

13. Idempotency (Exactly once):-

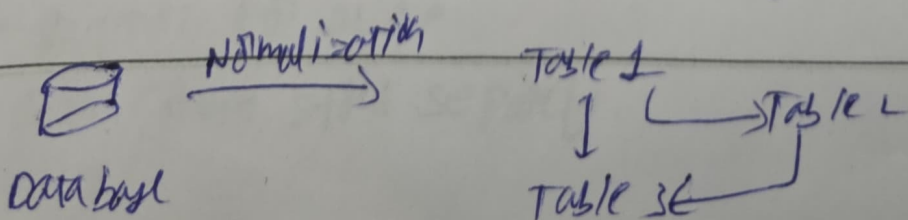
- * Data have to be processed at once.
- * We can achieve by using watermark approach.

14. Upsert:-

- * Update + Insert

15. Database modeling:- (OLTP)

- * It follows Normalization. to get good performance.
- * Inside database it store table in relation way by using modeling.
- 1NF $\rightarrow$ Primary key & No multi values.
- 2NF $\rightarrow$ No partial dependency
- 3NF $\rightarrow$ No transitive dependency.



16. Dimensional Data modeling - OLAP :-

* It follows Star & snowflake schemas.

17. Slowly changing dimensions :-

I know.

18. ORIT - One Big Table :-

* We combine ~~multiple tables~~ ^{multiple tables} ~~things~~ in a single table related to same domain.

19. Medallion Architecture :-

I know.

20. Batch processing :-

* To process data in batches

21. Stream processing :-

* To process data immediately.

22. Lambda Architecture :-

* Batch + Stream

23. Kappa Architecture:-

- * Stream.

24. CDC - Change data Capture:-

- * It focus on source side
- * It Detectes & extract data changes from source database.
- * It marked with operation type (Insert, update & delete)

25. Schema Evolution:-

I know

26. Row Based vs Column Based:-

Row Based:-

Ex:- CSV, JSON, AVRO, ORC

- * Better for OLTP
- * Data Skipping not possible.

Column Based:-

Ex:- Parquet

- * Better for OLAP
- * Each Column stored separately.

24. Distributed Data Processing:-

1. Parallelism

* It is like vertical scaling.

2. Distribution:-

* It is like Horizontal scaling.

25.